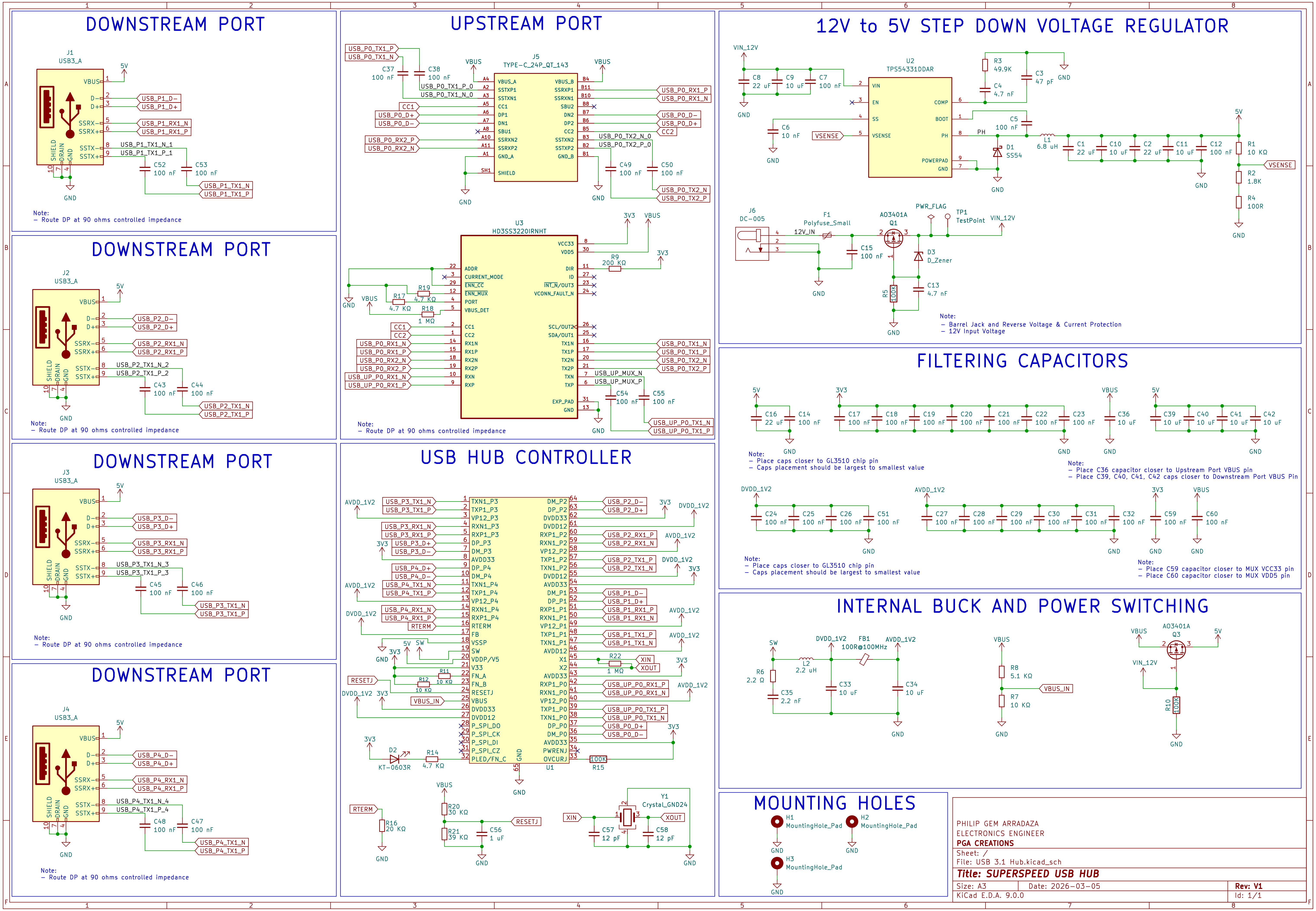

DOWNSTREAM PORT
J1 USB3_A
Note:
- Route DP at 90 ohms controlled impedance
DOWNSTREAM PORT
J2 USB3_A
Note:
- Route DP at 90 ohms controlled impedance
DOWNSTREAM PORT
J3 USB3_A
Note:
- Route DP at 90 ohms controlled impedance
DOWNSTREAM PORT
J4 USB3_A
Note:
- Route DP at 90 ohms controlled impedance
UPSTREAM PORT
J5 TYPE-C_24P_QT_143
U3 HD3SS3220IRNHT
Note:
- Route DP at 90 ohms controlled impedance
USB HUB CONTROLLER
U1
Y1 Crystal_GND24
12V to 5V STEP DOWN VOLTAGE REGULATOR
U2 TPS54331DDAR
J6 DC-005
F1 Polyfuse_Small
AO3401A Q1
Note:
- Barrel Jack and Reverse Voltage & Current Protection
- 12V Input Voltage
FILTERING CAPACITORS
Note:
- Place caps closer to GL3510 chip pin
- Caps placement should be largest to smallest value
Note:
- Place C36 capacitor closer to Upstream Port VBUS pin
- Place C39, C40, C41, C42 caps closer to Downstream Port VBUS Pin
Note:
- Place caps closer to GL3510 chip pin
- Caps placement should be largest to smallest value
Note:
- Place C59 capacitor closer to MUX VCC33 pin
- Place C60 capacitor closer to MUX VDD5 pin
INTERNAL BUCK AND POWER SWITCHING
AO3401A Q3
MOUNTING HOLES
H1 MountingHole_Pad
H2 MountingHole_Pad
H3 MountingHole_Pad
PHILIP GEM ARRADAZA
ELECTRONICS ENGINEER
PGA CREATIONS
Sheet: /
File: USB 3.1 Hub.kicad_sch
Title: SUPERSPEED USB HUB
Size: A3
Date: 2026-03-05
Rev: V1
KiCad E.D.A. 9.0.0
Id: 1/1